THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET A

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The legs of a right triangle are 3 cm and 4 cm. Its hypotenuse is: 1

(a) 5 cm **(b)** 4 cm **(c)** 6 cm **(d)** 7 cm

2. The legs of a right triangle are 20 cm and 21 cm. Its hypotenuse is: 1

(a) 41 cm **(b)** 29 cm **(c)** 28 cm **(d)** 30 cm

3. The hypotenuse of a right triangle is 5 cm and one leg is 3 cm. The other leg is: 1

(a) 4 cm **(b)** 2 cm **(c)** 5 cm **(d)** 3 cm

4. The hypotenuse of a right triangle is 29 cm and one leg is 20 cm. The other leg is: 1

(a) 20 cm **(b)** 22 cm **(c)** 9 cm **(d)** 21 cm

5. Is a triangle with sides 11, 60 and 61 a right triangle? 1

(a) Only if it is isosceles **(b)** No

(c) Yes **(d)** Cannot say

6. The diagonal of a square of side 4 cm is: 1

(a) 8 cm **(b)** $4\sqrt{2}$ cm **(c)** 4 cm **(d)** $4\sqrt{3}$ cm

7. The Baudhayana–Pythagoras theorem states that in a right triangle, the square on the hypotenuse equals: 1

(a) twice the product of the legs

(b) the difference of the squares

(c) the sum of squares on the other two sides

(d) the sum of the legs

8. The side of an isosceles right triangle is 5 cm. Its hypotenuse is: 1

(a) 10 cm **(b)** $5\sqrt{3}$ cm **(c)** $5\sqrt{2}$ cm **(d)** 7 cm

9. The triplet from $m = 3$, $n = 2$ is: 1

(a) (5, 12, 13) **(b)** (9, 12, 15) **(c)** (6, 8, 10) **(d)** (3, 4, 5)

10. The area of a square drawn on the hypotenuse of a right triangle with legs 6 and 8 is: 1

(a) 14 cm² **(b)** 100 cm² **(c)** 48 cm² **(d)** 196 cm²

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State the Baudhayana–Pythagoras theorem. 2

12. Find the hypotenuse of a right triangle with legs 9 cm and 12 cm. 2

13. Write the triplet that contains 12 and 35. 2

14. Is (10, 24, 26) a Pythagorean triplet? Verify. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A ladder of length 17 m rests against a wall with its foot 8 m from the wall. How high up the wall does it reach? 3

16. A ladder of length 25 m rests against a wall with its foot 7 m from the wall. How high up the wall does it reach? 3

17. A ladder of length 29 m rests against a wall with its foot 20 m from the wall. How high up the wall does it reach? 3

OR

A ladder of length 41 m rests against a wall with its foot 9 m from the wall. How high up the wall does it reach? 3

18. A ladder of length 10 m rests against a wall with its foot 6 m from the wall. How high up the wall does it reach? 3

OR

A ladder of length 20 m rests against a wall with its foot 12 m from the wall. How high up the wall does it reach? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) State the theorem and prove it using a square of side $a + b$ (outline). 5
(ii) Verify for the triangle with sides 7, 24, 25.

OR

Case study: A television screen has width 40 cm and height 30 cm. 5
(i) Find its diagonal.
(ii) A 50 cm rod: can it lie flat inside?
(iii) What diagonal would a screen of 24 cm × 32 cm have?

20. (i) Find the length of the altitude of an equilateral triangle of side 10 cm. 5
(ii) Hence find its area.

OR

(i) The sides of a right triangle are x, $x + 7$ and $x + 8$ (hypotenuse). Form an equation in x. 5
(ii) Find the sides.

END OF QUESTION PAPER — Total: 40 Marks

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